PROJECT DOCUMENTATION PHASE

Final Project Report • Tableau Public • Global Fuel Price Analysis

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Course	Bachelor of Computer Applications (BCA)	**Semester**	Second Year / 3rd Semester
College	MEDI-CAPS University	**Project type**	Individual Project

Global Fuel Prices and Taxation Analysis (2020–2026)

Purpose: To learn the concepts of Tableau and data analytics.

Executive Summary

This individual project uses Tableau Public to explore historical fuel prices, taxation, subsidy categories and economic groupings. The supplied CSV dataset contains approximately 27,468 records across 84 countries and 7 regions, covering January 2020 to April 2026. The workbook presents three scenario-based dashboards and a supporting historical trend worksheet. The report is descriptive and exploratory: it helps users compare patterns but does not establish causal relationships or provide live fuel-price monitoring.

Project Objectives

• Compare fuel-tax percentages and petrol prices across countries and regions.
• Explore fuel-price differences by subsidy level and income group.
• Compare high- and low-price countries and review petrol/diesel differences.
• Explore historical fuel-price movements alongside Brent crude values.
• Practise data connection, visualisation, dashboard design and publishing in Tableau.

Report Structure

Section	Coverage
1. Introduction & ideation	Problem context, purpose, analytical scenarios and project-defined personas.
2. Requirements & design	Dataset, tools, user needs, dashboard plan and solution architecture.
3. Planning & development	Backlog, illustrative sprint plan, worksheets and three dashboards.
4. Testing & results	Evidence-based functional observations, dashboard outputs and limitations.
5. Conclusion & future scope	Benefits, constraints, possible extensions and project links.

Scope note: The project uses the supplied CSV connected directly to Tableau. A separate SQL database, automated live feed, independent ETL pipeline and formal load-testing results are not part of the confirmed implementation.

1. Requirements, Data & Solution Design

Problem Statement

A large table of country-level fuel prices, taxes, subsidies and dates is difficult to compare quickly. The project organises selected measures into labelled charts and dashboards so a viewer can explore regional differences, country rankings, subsidy patterns and income-level comparisons.

Dataset & Technology

Item	Details
Dataset	CSV containing date, country, region, income_level, subsidy_level, petrol_usd_liter, diesel_usd_liter, lpg_usd_liter, brent_crude_usd and tax_percentage.
Coverage	Approximately 27,468 records; 84 countries; 7 regions; January 2020–April 2026 (as described in the supplied project notes).
Analysis platform	Tableau Public for data connection, worksheets, visualisations, dashboard arrangement, filters/legends and publication.
Data flow	CSV file → Tableau data source → worksheets and aggregations → three dashboards → published Tableau Public workbook.

Dashboard Specifications

Dashboard	Main analytical views	Purpose
1. Fuel Taxation & Subsidy Policy	Top countries by average tax percentage; average petrol price by region; taxation versus petrol-price scatter plot; fuel prices by subsidy level.	Explore policy-related comparisons and tax/subsidy patterns.
2. Global Fuel Prices & Supply Chain Planning	Top and bottom 10 countries by average petrol price; petrol/diesel comparison; regional price comparison; Brent crude and fuel-price trend view.	Compare country price levels and historical patterns relevant to planning.
3. Fuel Prices, Taxation & Economic Development	Average petrol price and average tax percentage by income level; taxation/price and subsidy comparisons; visible Income Level, Region and Subsidy Level filters.	Explore differences across income groups and categories.

Interpretation rule: Visual associations between taxation, prices, subsidy levels, income groups and Brent crude should be treated as descriptive patterns—not proof that one factor caused another.

2. Development, Testing & Results

Development Summary

The workbook was developed by connecting the supplied CSV to Tableau, creating supporting worksheets, arranging charts into three scenario dashboards, adding relevant legends and visible filters, and publishing the workbook to Tableau Public. The supporting worksheet “Global Petrol Price Trend” is not counted as a fourth dashboard.

Functional & Performance Testing

Area	Evidence / result	Limit
Workbook & dashboard structure	Published workbook and three scenario dashboards are documented; each dashboard is planned around four visualisations.	Screenshots show structure; not a formal usability study.
Filters and legends	Scenario 3 shows Income Level, Region and Subsidy Level controls; relevant dashboards include legends.	Filter latency and usability scores were not measured.
Website checks	The supplied testing reference reports four basic website functional checks as passed: availability, page content, embedded dashboard and interaction.	These are reported functional checks, not performance benchmarks.
Load, concurrency & mobile	No measured rendering-time or concurrency result is available.	Not formally tested; avoid numerical speed claims.

Results & Benefits

The main result is a published Tableau Public workbook that presents three scenario dashboards for policy analysis, supply-chain planning and economic-development exploration. The visual approach makes it easier to compare countries, regions, income groups and subsidy categories than reading the raw dataset alone.

Advantages

- Visual comparisons help surface patterns in a large dataset.
- Scenario-based dashboards give the analysis a clear purpose.
- Published Tableau workbook is accessible through a browser.
- Filters and legends help viewers focus on selected categories.

Limitations

- Results depend on the coverage and quality of the supplied historical dataset.
- Aggregated averages can hide differences within countries or groups.
- The workbook is not a live market-monitoring service.
- Visual correlation does not establish causation; performance and mobile behaviour were not formally benchmarked.

3. Conclusion, Future Scope & Appendix

Conclusion

The project demonstrates practical use of Tableau for exploratory data analysis. By organising historical fuel prices, tax percentages, subsidy levels and income categories into three dashboards, it provides a structured way to examine cross-country and regional patterns. The project's learning objective is to build familiarity with data connection, visualisation, dashboard design, interpretation and publication—not to provide causal economic conclusions or real-time price forecasts.

Future Scope

- Add documented date and category filters where useful.
- Conduct a structured review of chart readability and dashboard navigation.
- Measure page-load and dashboard-rendering times using a repeatable method.
- Test layouts on mobile viewport sizes.
- Consider a scheduled data refresh only if a reliable, permitted source becomes available.
- Add a separate database or automated data pipeline only if the project's scope expands.

Project Links

Tableau Public Story	Open Disha's published Tableau project
Project website	global-fuel-prices-taxation-analysis.onrender.com
GitHub repository	DishaPatidar11/global-fuel-prices-taxation-analysis

PROJECT PURPOSE	To learn the concepts of Tableau and data analytics through scenario-based analysis of global fuel prices, taxation and subsidy patterns.